

SHORT HORIZON REINFORCEMENT LEARNING

Odalric-Ambrym Maillard
odalric.maillard@inria.fr
Centre Inria de l'Université de
Lille,
Villeneuve d'Ascq, FRANCE

TBD
TBD
TBD,
TBD

TBD
TBD
TBD,
TBD

Abstract

[\[Link to inria notes\]](#)

[\[Sharing link\]](#)

1 Introduction

Regret lower bounds: [1, 6]. Extensions: [5, 4, 7, 11].

Setup: regret minimization, best-arm, or alternative (e.g. mixed).

Learner assumption: consistent, or more (symmetric, etc).

Asymptotic lower bounds for consistent learners.

Comparison of lower bounds.

KL-UCB-Switch [5, Th.2]: For all $T \geq T_0(\nu)$, $\mathbb{E}[N_t(a)] \leq \frac{\ln(T) - \ln \ln(T)}{\mathcal{K}_{\inf}(\nu_a, \mu^\star)} + B$

1.1 Regret minimization

In this section, we focus exclusively on the minimization of the cumulative regret objective. Resorting to the regret lower bounds and highlighting the regret performance achievable by best strategies, we revisit this notion, from the perspective of the practitioner acting in short time horizons, and introduce the **Practitioner regret** that informally speaking penalizes a learning agent only with respect to what could have been achieved by a best performing learning agent in the same context. We now detail and formalize this notion.

Notations Let $\mathbf{m}^\star = \max_a \mathbf{m}(\nu_a)$ denote the maximal mean reward of environment ν . Let us introduce $\Delta^\varepsilon(a) = (\mathbf{m}^\star - \varepsilon - \mathbf{m}(\nu_a))_+$ the ε -optimality gap of arm a (where $(x)_+ = \max(x, 0)$), and $O_\varepsilon = \{a \in \mathcal{A} : \mathbf{m}(\nu_a) \geq \mathbf{m}^\star - \varepsilon\} = \{a \in \mathcal{A} : \Delta^\varepsilon(a) = 0\}$ be the set of ε -near-optimal arms.

Near-optimality The ε -optimal regret of algorithm $\mathbb{A}$ interacting with ν for T time steps writes $\mathcal{R}_T^\varepsilon(\mathbb{A}, \nu) = \sum_{t=1}^T \mathbb{E}[(\mathbf{m}^\star - \varepsilon - \mathbf{m}(\nu_{a_t}))_+] = \sum_{a \notin O_\varepsilon} \mathbb{E}[N_t(a)] \Delta^\varepsilon(a)$. The corresponding regret lower bound writes, for all reasonable enough, that is, consistent, algorithm $\mathbb{A}$,

$$\liminf_{T \rightarrow \infty} \frac{\mathcal{R}_T^\varepsilon(\mathbb{A}, \nu)}{\log(T)} \geq C^\varepsilon(\nu) = \sum_{a \in \mathcal{A}} C_a^\varepsilon(\nu) \text{ where } C_a^\varepsilon(\nu) = \frac{\Delta^\varepsilon(a)}{\mathcal{K}_{\inf}(\nu_a, \mu^\star - \varepsilon)}.$$

Such lowerbounds are usually more precise, showing that actually

$$\liminf_{T \rightarrow \infty} \frac{\mathbb{E}[N_T(a)]}{\log(T)} \geq C_a^\varepsilon(\nu).$$

OM: We can further study non-asymptotic lower bounds, under different assumptions about the learner (than consistency, which is asymptotic).

Regime-separation critical times Noting that, trivially, $\mathbb{E}[N_t(a)]\Delta^\varepsilon(a) \leq T\Delta^\varepsilon(a)$, and that, asymptotically, $\mathbb{E}[N_T(a)] \geq C_a^\varepsilon(\nu) \log(T)$, we introduce for each arm a and ε the critical time $\tau^\varepsilon(a) = \inf \left\{ T : \frac{1}{\mathcal{K}_{\inf}(\nu_a, \mu^\star - \varepsilon)} < \frac{T}{\log(T)} \right\}$, that is the smallest time T for which the contribution of arm a to the logarithmic regret bounds becomes non trivial. We further introduce for $\eta \in (0, 1]$

$$\tau_\eta^\varepsilon(a) = \inf \left\{ T : \frac{1}{\mathcal{K}_{\inf}(\nu_a, \mu^\star - \varepsilon)} < \eta \frac{T}{\log(T)} \right\},$$

imposing a stronger separation between the asymptotic and the non-asymptotic regime. Note that if $\eta \leq \eta' \leq 1$, then $\tau_1^\varepsilon(a) \leq \tau_{\eta'}^\varepsilon(a) \leq \tau_\eta^\varepsilon(a)$, requiring larger asymptotic times as η decreases. Likewise, if $0 \leq \varepsilon \leq \varepsilon'$, then $\tau_\eta^\varepsilon(a) \leq \tau_\eta^{\varepsilon'}(a) \leq \tau_\eta^0(a)$.

OM: More generally, for any lower bound expressed as $\mathbb{E}[N_t(a)] \geq L_t(a, \nu, \varepsilon)$ for all t (or all for all t sufficiently large, $t \geq T_0$) we define $\tau_\eta^\varepsilon(a) = \inf \{ T : L_T(a, \nu, \varepsilon) < \eta T \}$, corresponding to the first time when the lower bound becomes non-trivial.

Practitioner Regret Fixing the separation factor η , we define $\varepsilon_\eta(T, \nu) = \inf \{ \varepsilon : T \geq \max_a \tau_\eta^\varepsilon(a) \}$ to be the smallest ε such that the asymptotic regime appears simultaneously for all arms (with η separation). We are now ready to introduce a novel notion of regret, which we call the **Practitioner regret**. It is defined for a learning agent $\mathbb{A}$ having interacted with environment ν for T time steps by $\mathcal{R}_T^{\eta, \dagger}(\mathbb{A}, \nu) = \mathcal{R}_T^{\varepsilon_\eta(T, \nu)}(\mathbb{A}, \nu)$.

Interpretation of the practitioner regret Note that for a given ν , as $T \rightarrow \infty$, $\varepsilon_\eta(T, \nu) \rightarrow 0$ hence $\mathcal{R}_T^{\eta, \dagger}(\mathbb{A}, \nu) \simeq \mathcal{R}_T^0(\mathbb{A}, \nu)$ asymptotically. That is the practitioner regret asymptotically behaves like the classical regret, which is a desirable property. Now the practitioner regret may differ significantly from both the regret or near-optimal regret in *short* time horizons. For instance when $\varepsilon_\eta(T, \nu) \geq \max_a \Delta_a$, which may happen for very small time horizons $T < A$ and appropriate environment ν , then $\mathcal{R}_T^{\eta, \dagger}(\mathbb{A}, \nu) = 0$ for all $\mathbb{A}$ that is no regret is considered to be incurred by any algorithm $\mathbb{A}$ in this regime, while in the same regime, only $\mathcal{R}_T^0(\mathbb{A}, \nu) \leq T \max_a \Delta_a$ seems achievable. The reason for this difference is that the practitioner regret does not score the performance of the agent to the best absolute precision level ($\varepsilon = 0$), but rather to the best precision one can effectively target at a given time for this environment ($\varepsilon_\eta(T, \nu)$), ensuring the regret at this precision has indeed enters its non-trivial asymptotic regime. In stark contrast, for short time horizons, the classical notions of regret $\mathcal{R}_T(\mathbb{A}, \nu)$ and ε -optimal regret $\mathcal{R}_T^\varepsilon(\mathbb{A}, \nu)$ both overly penalize an algorithm for an error that cannot be controlled even by a best performing algorithm.

Hence, assuming the regret lower bounds are tight (there exists an algorithm achieving it), we consider this notion of regret to be more meaningful for the practitioner, since it captures what could be effectively captured by a best performing algorithm. For this reason we call $\mathcal{R}_T^{\eta, \dagger}(\mathbb{A}, \nu)$ the **Practitioner regret** of a learning agent in environment ν .

Bounding the practitioner regret Since $\mathcal{R}_T^{\eta, \dagger}(\mathbb{A}, \nu) \leq \mathcal{R}_T^0(\mathbb{A}, \nu)$, any bound of the classical regret transfers immediately to a control of the practitioner regret. One may then ask what is the benefit of studying this concept. We will see that this becomes interesting in the regimes when the usual bound offer a bad scaling with parameters of the problem. For instance, in the MDP or Markov arm setting, it is natural to see the regret bounds scaling with $1/p_{\min}$ where $p_{\min}$ is the minimal non-zero transition probability between two states (or closely related quantities such as the Diameter of the MDP). This becomes a major concern when $p_{\min}$ is small, so that $1/p_{\min}$ is much higher than any practical time horizon T , say $> 10^{15}$. At the bounds crucially depend on such parameters, they become meaningless for the practitioner. In such cases however, we claim the regret should be amended, since no reasonable algorithm can achieve absolute regret optimality for shorter horizon. It then since natural to consider a more lenient notion, penalizing the algorithm for what is actually

achievable. The aim is then to show that a bound on the practitioner regret can be provided, without scaling with $1/p_{\min}$.

OM: This is of course a conjecture at this point.

Best-arm identification We now proceed to provide the lower performance bounds for best-arm identifications it holds

$$\liminf_{\delta \rightarrow 0} \frac{\mathbb{E}[\tau_\delta]}{\text{kl}(\delta, 1 - \delta)} \geq \mathcal{D}(\nu) = \frac{1}{\sup_{\omega \in \Delta_{\mathcal{A}}} \inf_{\nu' \in \text{Alt}(\nu)} \sum_{a \in \mathcal{A}} \omega_a \text{KL}(\nu_a, \nu'_a)}$$

$$\sum_a n_a \text{ s.t. for each } a, \inf_{\nu' \in \text{Alt}(\nu, a)} \sum_{a' \in \mathcal{A}} n_{a'} \text{KL}(\nu_{a'}, \nu'_{a'}) \geq \text{kl}(\delta, 1 - \delta)$$

When considering the top- k objective instead, we get instead...

OM: Should read Tianyuan Jin's other works actually: they are actually extending the Double-KL-UCB paper to ε -optimality and top k .

2 Lower bounds literature

[8]

[2]

[5]

[3]

[10] Peter thinks the impossibility results does not hold

[9] Ronald does not understand this paper, maybe we can discuss it.

OM: Same: Actually if you weaken assumptions on the learner (only Hannan consistent), but keep a class of distributions that concentrates enough (e.g. sub-Gaussian), then I also believe you can get non trivial lower bounds. I wrote some doc about this actually.

3 Preliminary Experiments

A few experiments to be done to build intuition:

- Todo1: For a given ν , compute/plot $T \rightarrow \varepsilon(T, \nu)$
- Todo2: Compute the corresponding practitioner regret of UCB/TS/KLUCB
- Todo3: From observations, compute some estimate $\widehat{\varepsilon}(t, \nu)$ at time t . Try heuristic idea: Lenient UCB/KLUB/TS with $\widehat{\varepsilon}(t, \nu)$ (vs arbitrary leniency)

References

- [1] Rajeev Agrawal, Demosthenis Teneketzis, and Venkatachalam Anantharam. Asymptotically efficient adaptive allocation schemes for controlled i. i. d. processes: Finite parameter space. *IEEE Transactions on Automatic Control*, 34(3):258–267, 1989.
- [2] Peter Auer, Nicolò Cesa-Bianchi, Yoav Freund, and Robert E. Schapire. The nonstochastic multiarmed bandit problem. *SIAM Journal of Computing*, 32(1):48–77, 2002.
- [3] Sébastien Bubeck, Vianney Perchet, and Philippe Rigollet. Bounded regret in stochastic multi-armed bandits. 2013.

- [4] Aurélien Garivier, Hédi Hadiji, Pierre Menard, and Gilles Stoltz. Kl-ucb-switch: optimal regret bounds for stochastic bandits from both a distribution-dependent and a distribution-free viewpoints. *Journal of Machine Learning Research*, 23(179):1–66, 2022.
- [5] Aurélien Garivier, Pierre Ménard, and Gilles Stoltz. Explore first, exploit next: The true shape of regret in bandit problems. *Mathematics of Operations Research*, 44(2):377–399, 2019.
- [6] Todd L Graves and Tze Leung Lai. Asymptotically efficient adaptive choice of control laws in controlled markov chains. *SIAM journal on control and optimization*, 35(3):715–743, 1997.
- [7] Cyrille Kone, Emilie Kaufmann, and Laura Richert. Adaptive algorithms for relaxed pareto set identification. *Advances in Neural Information Processing Systems*, 36:35190–35201, 2023.
- [8] Tze Leung Lai and Herbert Robbins. Asymptotically efficient adaptive allocation rules. *Advances in Applied Mathematics*, 6:4–22, 1985.
- [9] Tor Lattimore. The pareto regret frontier for bandits. In *Advances in Neural Information Processing Systems 28*, pages 208–216, 2015.
- [10] Antoine Salomon and Jean-Yves Audibert. Deviations of stochastic bandit regret. In Jyrki Kivinen, Csaba Szepesvári, Esko Ukkonen, and Thomas Zeugmann, editors, *Algorithmic Learning Theory - 22nd International Conference, ALT 2011*, volume 6925 of *Lecture Notes in Computer Science*, pages 159–173. Springer, 2011.
- [11] Junwen Yang, Vincent YF Tan, and Tianyuan Jin. Best arm identification with minimal regret. *arXiv preprint arXiv:2409.18909*, 2024.